

TAMIL NADU STATE BOARD - STANDARD 11 PHYSICS VOLUME 2

CHAPTER 7: PROPERTIES OF MATTER

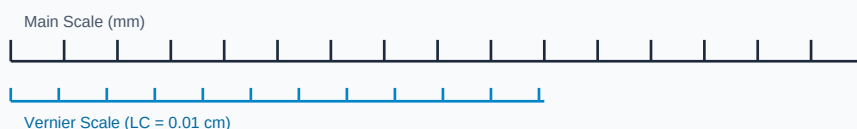
PREMIUM ACADEMIC STUDY SUITE & EXAM GUIDE

Board:	Tamil Nadu State Board (TNSB)
Syllabus:	Samacheer Kalvi Textbook Curriculum aligned
Standard:	Class 11 English Medium
Subject:	Physics Volume 2
Chapter Name:	Properties of Matter

Learning Objectives:

- Comprehend core theoretical concepts and exact textbook terminology.
- Apply relevant formulas and proofs to solve high-scoring board problems.
- Interpret and sketch vector diagrams or graphical models representing equations.
- Build examination readiness for 1-mark, 2-mark, and 5-mark question sets.

Measurement Linear Scale Division



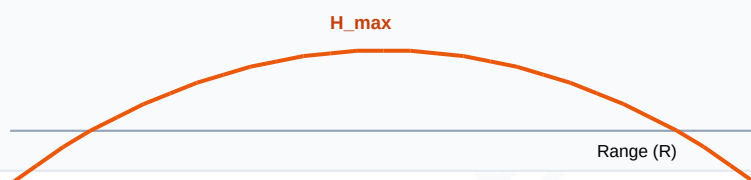
Core Concepts & Definitions

Textbook definitions are essential for scoring fully in the short-answer section of board assessments. Review the primary definitions below:

Term	Official Definition & Context
Stress and Strain	Stress is internal restoring force per unit area; strain is fractional change in dimension under load.
Surface Tension	The property of a liquid surface to act like a stretched elastic membrane, minimizing area.
Viscosity	The internal friction force between fluid layers moving relative to each other.

Syllabus Exam Tip: Memorize precise keywords. Examiners award maximum marks when textbook terms and correct symbols are correctly referenced.

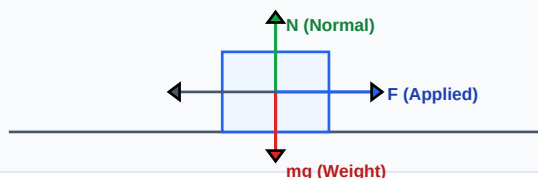
Kinematic Projectile Trajectory



Detailed Explanation & Formulas

Here, we analyze the core laws and formulas. Examine the conceptual illustration below to visual the relation between variables:

Force Vector Free Body Diagram (FBD)



Essential Formulas, Laws & Identities:

- Hooke's Law: Stress = $E \times \text{Strain}$ | Young Modulus: $Y = (F \cdot L) / (A \cdot \Delta L)$
- Terminal Velocity (Stokes): $v_t = (2/9) \cdot r^2 \cdot (\rho_s - \rho_f) \cdot g / \eta$
- Surface Tension: $T = F / l = r \cdot h \cdot \rho \cdot g / (2 \cdot \cos(\theta))$

Make sure to check the dimensional consistency of all physical quantities and verify that units are correctly stated in final solutions.

Worked Examples

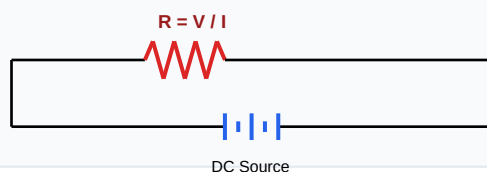
Review these standard board examination-style worked examples to learn proper step layout structures:

Example 1: Find terminal velocity of a steel ball of radius 1mm falling in glycerine ($\eta=0.83 \text{ kg/m-s}$, densities steel=7800, glycerine=1260 kg/m^3).

Step-by-step Solution:

$$v_t = \frac{2}{9} (10^{-3})^2 \cdot (7800 - 1260) \cdot 9.8 / 0.83 = \frac{2}{9} \cdot 10^{-6} \cdot 6540 \cdot 9.8 / 0.83 = 0.017 \text{ m/s}.$$

Schematic Electric Circuit Diagram



HOTS (Higher Order Thinking Skills) Challenge

Challenge yourself with these complex, multi-concept problems designed to test deeper conceptual understanding and mathematical reasoning:

HOTS Challenge 1: State and prove Bernoulli's Theorem for streamline fluid flow.

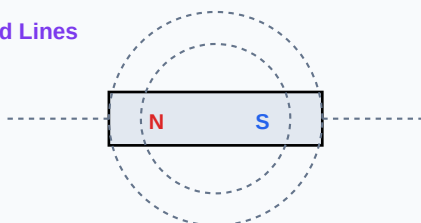
Analytical Solution Strategy:

Theorem: $P + 0.5\rho v^2 + \rho gh = \text{Constant}$.

Proof: Work done per unit volume by pressure equals changes in KE and PE per unit volume.

$P_1 - P_2 = 0.5\rho(v_2^2 - v_1^2) + \rho g(h_2 - h_1)$, rearranges to standard form.

Magnetic Dipole Bar Field Lines



Practice Worksheet

Test your skills by attempting these sample exam problems independently. Draw diagrams where appropriate:

Q1 (1-Mark): State Hooke's Law and draw the stress-strain curve.

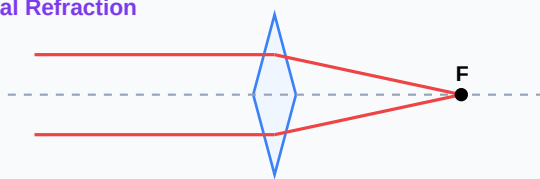
Q2 (2-Mark): Explain terminal velocity concept and state its expression.

Q3 (2-Mark): What is capillarity? Write down the capillary rise formula.

Q4 (5-Mark): State Pascal's Law and list two applications.

Q5 (5-Mark): Define Reynolds number and state its significance.

Convex Lens Optical Refraction



Real-World Applications & Connections

This chapter connects directly with modern industrial engineering, computation, and natural systems in numerous ways:

Application Case: Civil Construction

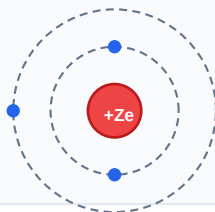
I-beams are designed using stress calculations to carry heavy loads without bending.

Application Case: Inkjet Printers

Ink droplets are formed using capillary pressure control in nozzle structures.

Cross-Curricular Link: Concepts learned in this unit are heavily applied in other subjects like Physics (equations of motion, field vectors) and Data Analytics.

Bohr Atomic Shell Configuration



Revision Summary & Strategy

Use this checklist to self-evaluate your exam preparation status before final tests:

Key Recall Tips:

- ✓ Always write down 'Given Parameters' first to ensure partial marks are locked in.
- ✓ Pay attention to signs (+ / -) when performing algebraic operations or force mappings.
- ✓ Check that you have written the correct units (e.g. Joules, Tesla, cm^3 , N) for the final numeric answer.
- ✓ Draw neat diagrams using a sharp pencil and ruler for geometry or circuit layouts.

Syllabus Checklist:

- ☐ Read and memorized core definitions on Page 2.
- ☐ Reviewed critical formulas and relations on Page 3.
- ☐ Solved worked examples on Page 4.
- ☐ Attempted worksheet problems on Page 6.

CONGRATULATIONS! YOU HAVE COMPLETED THIS SYLLABUS STUDY GUIDE!

Practice consistently, verify answers with standard textbook methods, and take the topic test in the Nexora Learning app to gauge progress.
Good luck!

Sinusoidal AC Voltage Waveform

